Bojian HOU, Ph.D.

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PROFESSIONAL SUMMARY

- AI Research Scientist at Meta with 12+ years in AI research and 6+ years of post-PhD experience across industry and academia, spanning large-scale recommendation, large language models, and Trustworthy AI (calibration, alignment, fairness)
- Leading research since 2022: at Penn, a Trustworthy AI research line with up to 10 students, 6+ concurrent projects, and 5+ papers as corresponding author; at Meta, first author of Kunlun (KDD'26), an LLM-style ads ranking architecture deployed in major Meta Ads models (MFU 17% to 37%), and owner of the O2A project across 5+ teams and five launch paths
- Published 50+ papers with 1,000+ Google Scholar citations at NeurIPS, ICML, ICLR, COLM, KDD, EMNLP, AAAI, UAI, TKDE, TNNLS, and more, including recent LLM calibration and alignment work (NeurIPS'26, COLM'26, ICML'25)
- Recognized with the *Distinguished Paper Award* (AMIA'24), *Best Paper Award* (ACM BCB'23, BCB'25), *PennAITech Innovation Fellow Award*, *Outstanding Student Paper Award* (CCFAI'19), and *Excellent Doctoral Dissertation Awards* (Jiangsu, JSAI, NJU)

EDUCATION

Nanjing University, Computer Science, *Ph.D.* (GPA: 4.46/5.00, 3/60), Nanjing, China — Sep. 2014 - Jun. 2020

- Expertise in machine learning and artificial intelligence, LAMDA Group (advisor: Prof. Zhi-Hua Zhou)

Nanjing University, Computer Science, *Bachelor of Science* (GPA: 4.33/5.00, 20/180), Nanjing, China — Sep. 2010 - Jun. 2014

WORK EXPERIENCE

Meta, AI Research Scientist, Monetization & Meta Recommendation Systems (MRS), Menlo Park, CA — Jul. 2025 - Present

- First-authored Kunlun (KDD'26) and led its framing, writing, and figures for a ~30-author Meta Ads effort on an LLM-style unified architecture for late-stage ads ranking with predictable scaling laws; the architecture raises MFU from 17% to 37% on NVIDIA B200 GPUs, doubles scaling efficiency over state-of-the-art methods, and is deployed in major Meta Ads models
- Revived the organic-to-ads (O2A) representation project after a reorg left it with no offline gain and most contributors gone: introduced impressed-but-unclicked ads as hard negatives with an ordinal floor loss, improving offline top-1 accuracy by ~30%
- Own O2A end to end across 5+ partner teams (including data engineering and the Facebook recommendation product team): PyTorch contrastive modeling, Hive/Presto embedding pipelines, online pre-tests, and five launch paths spanning user-to-user and ad-to-ad retrieval, ads boosting, feature injection, and sequence filtering
- Designed a request-only (RO) architecture for the production late-stage ranking model that computes request-level components once per request instead of once per candidate, delivering a 0.05% NE gain with no QPS regression
- Led a research project on Directed Upstream Learning, which adapts upstream models for frozen downstream recommenders, supervising a PhD research intern from method design and theory through a WSDM'27 submission
- Published LLM calibration research as co-corresponding author: bilevel fine-tuning that keeps aligned LLMs calibrated under domain shift where post-hoc temperature scaling fails (COLM'26), and a semantic-sampling calibration metric for open-ended QA with provably unbiased estimators, validated on five frontier LLMs (NeurIPS'26)

University of Pennsylvania, Postdoctoral Fellow, Philadelphia, PA — Jan. 2022 - Jul. 2025

- Led a Trustworthy AI research line (LLM calibration, fairness-aware and interpretable machine learning, vision-language models) with applications to Alzheimer's disease and mental health, running 6+ projects at a time
- Mentored 20+ students (up to 10 at a time) through the full research cycle (problem formulation, coding, experiments, compute setup, paper writing, and rebuttals), resulting in co-authored papers at NeurIPS, ICML, ICLR, and KDD
- Showed that preference alignment degrades LLM calibration and developed calibration-aware fine-tuning (ICML'25); proposed attention-only fine-tuning that improves weight disentanglement for model merging via task arithmetic (ICLR'25)
- Built MentalChat16K, a counseling benchmark for LLMs (KDD'25 Oral); developed SEFD for paraphrase-robust detection of LLM-generated text (IEEE BigData'24) and DALK for LLM and knowledge-graph co-augmentation (EMNLP'24 Findings)
- Developed fairness-aware learning methods: Fair CCA (NeurIPS'23), fairness-aware graphical models (NeurIPS'24), and fairness-aware class-imbalanced learning (UAI'23); built interpretable deep clustering survival machines (MedIA'24)

Cornell University, Postdoctoral Associate, New York, NY — May 2021 - Jan. 2022

- Developed a BERT-based clinical decision support system with interdisciplinary clinical teams, improving literature retrieval accuracy by 4% and information extraction F1 by 5%; studied Shapley value stability for interpretability of medical AI and used contrastive learning to improve the reliability of clinical decision support tools

4Paradigm Corporation, Research Scientist, Beijing, China — Aug. 2020 - Apr. 2021

- Pioneered "Environment Learning" for decision optimization, combining data-driven and knowledge-based approaches with reinforcement learning; virtual environment models cut validation loss by 79% for COVID-19 infection prediction

SELECTED PUBLICATIONS (1,000+ Google Scholar citations, * indicates co-first author, † indicates corresponding author)

[KDD'26] **Bojian Hou**, Xiaolong Liu, Xiaoyi Liu, Jiaqi Xu, Yasmine Badr, Mengyue Hang, et al. Kunlun: Establishing Scaling Laws for Massive-Scale Recommendation Systems through Unified Architecture Design. In: Proceedings of the 32nd ACM SIGKDD Conference on Knowledge Discovery and Data Mining (ADS Track), 2026.

- Unified architecture with predictable power-law scaling, combining GDPA, hierarchical seed pooling, sliding-window attention, computation skip, and event-level personalization; deployed in major Meta Ads models

[NeurIPS'26] Zhanliang Wang*, Jiancong Xiao*, Ruochen Jin, Shu Yang, **Bojian Hou**†, and Li Shen†. A Semantic-Sampling Framework for Evaluating Calibration in Open-Ended Question Answering. In: Proceedings of the 40th Annual Conference on Neural Information Processing Systems (NeurIPS 2026), 2026, in press. arXiv:2605.08432.

- Sem-ECE: logit-free calibration metric for open-ended QA with unbiased estimators, validated on five frontier LLMs; the gap between same-sample and held-out estimators also diagnoses question difficulty

[COLM'26] Ruochen Jin*, Zhanliang Wang*, Zongyu Dai, Jiancong Xiao†, and **Bojian Hou**†. Beyond Post-Hoc Temperature Scaling: Bilevel Optimization for LLM Calibration. In: Proceedings of the Conference on Language Modeling (COLM), 2026.

- CALM: first-order bilevel fine-tuning that keeps aligned LLMs calibrated under domain shift

[ICML'25] Jiancong Xiao*, **Bojian Hou***, Zhanliang Wang*, Ruochen Jin, Qi Long, Weijie J. Su, and Li Shen. Restoring Calibration for Aligned Large Language Models: A Calibration-Aware Fine-Tuning Approach. In: Proceedings of the 42nd International Conference on Machine Learning (ICML), Vancouver, Canada, 2025.

- Showed preference alignment degrades LLM calibration; calibration-regularized fine-tuning restores it without hurting accuracy

[ICLR'25] Ruochen Jin*, **Bojian Hou***, Jiancong Xiao*, Weijie Su, and Li Shen. Fine-Tuning Attention Modules Only: Enhancing Weight Disentanglement in Task Arithmetic. In: Proceedings of the 13th International Conference on Learning Representations, 2025.

- Fine-tuning only attention modules improves weight disentanglement and model merging via task arithmetic

[KDD'25] Jia Xu*, Tianyi Wei*, **Bojian Hou***, Patryk Orzechowski, Shu Yang, Ruochen Jin, Rachael Paulbeck, Joost Wagenaar, George Demiris, and Li Shen. MentalChat16K: A Benchmark Dataset for Conversational Mental Health Assistance. In: Proceedings of the 31st ACM SIGKDD Conference on Knowledge Discovery and Data Mining, 2025. (Oral)

- Benchmark of synthetic and anonymized real counseling dialogues for fine-tuning and evaluating LLM mental-health assistants

[NeurIPS'25] Parvin Nazari, **Bojian Hou**, Davoud Ataee Tarzanagh, Li Shen, and George Michailidis. Stochastic Regret Guarantees for Online Zeroth- and First-Order Bilevel Optimization. In: Advances in Neural Information Processing Systems 38, 2025.

- Regret guarantees for stochastic online bilevel optimization with zeroth- and first-order methods

[NeurIPS'24] Zhuoping Zhou*, Davoud A. Tarzanagh*, **Bojian Hou***, Qi Long, and Li Shen. Fairness-Aware Estimation of Graphical Models. In: Advances in Neural Information Processing Systems 37, 2024.

- Fair graphical model estimation that reduces bias across protected groups while preserving model quality

[NeurIPS'23] Zhuoping Zhou*, Davoud A. Tarzanagh*, **Bojian Hou***, Boning Tong, Jia Xu, Yanbo Feng, Qi Long, and Li Shen. Fair Canonical Correlation Analysis. In: Advances in Neural Information Processing Systems 36, 2023.

- Learns fair global projections by minimizing correlation disparity across protected groups while maintaining accuracy

[IEEE BigData'24] Weiqing He*, **Bojian Hou***, Tianqi Shang*, Davoud A. Tarzanagh, Qi Long, and Li Shen. SEFD: Semantic-Enhanced Framework for Detecting LLM-Generated Text. In: 2024 IEEE International Conference on Big Data, 2024.

- Retrieval-based semantic framework that makes LLM-generated text detection robust to paraphrasing attacks

[UAI'23] Davoud A. Tarzanagh*, **Bojian Hou***, Boning Tong, Qi Long, and Li Shen. Fairness-Aware Class Imbalanced Learning on Multiple Subgroups. In: Proceedings of the 39th Conference on Uncertainty in Artificial Intelligence, 2023, 2123-2133.

- Tri-level optimization for overparameterized models that improves generalization on imbalanced subgroups

[MLMI'26] Rahul Shukla, Ruiming Wu, Hongzhuo Chen, Shu Yang, Duy Duong-Tran, et al., **Bojian Hou**†, and Li Shen†. A Unified 3D Vision-Language Framework for Structured Multi-Study Volumetric Report Generation. In: Proceedings of the 17th International Conference on Machine Learning in Medical Imaging (MLMI), 2026, in press.

- Native 3D vision-language model for multi-study CT/MRI report generation, outperforming 2D and 3D medical VLMs

Earlier first-author work on feature evolvable streams and interpretable RNNs: NeurIPS'17, IJCAI'17, TKDE'19, TNNLS'20/'21, AAAI'21.

TECHNICAL SKILLS

Methods: LLM fine-tuning, preference optimization and calibration, LLM-as-judge evaluation, attention architectures and scaling for ranking, contrastive representation learning with hard negatives, fairness, interpretability, bilevel optimization
Programming Languages: Python, Java, C/C++, SQL, MATLAB, Bash, LaTeX, HTML, CSS
Machine Learning Platform/Tools: PyTorch, FBLearner Flow, Hugging Face, TensorFlow, Scikit-Learn, Pandas, Hive, Presto

ACADEMIC ACTIVITIES

(Senior) Program Committee of NeurIPS (2018-), ICML, ICLR, AAAI, AISTATS (2019-), KDD (2020-), and IJCAI (2021-)
Reviewer of Nature Methods, TPAMI, TNNLS, TII, Medical Image Analysis, Machine Learning, TKDD, and Scientific Reports